GROUP 37 · FINAL PRESENTATION

WildGuard

Learning from US wildlife enforcement to rank global CITES trade for human review

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NUS-ISS Graduate Certificate in Pattern Recognition Systems · Practice Module

THE PROBLEM

More trade than anyone can inspect

29.2 M

records in the CITES Trade Database since 1975 (version 2026.1)

~800,000

animal-trade records a year (668,000–848,000 in 2018–2023)

1.7%

of US wildlife import line items drew enforcement action (LEMIS, 2000–2014)

Inspection has to be targeted, and the risk signal is rarely one field: it sits in **combinations** of species, product, quantity, purpose, source and route, judged against what is normal for that species.

THE IDEA

LEMIS has the outcomes; CITES has the scale

LEMIS · US Fish & Wildlife Service

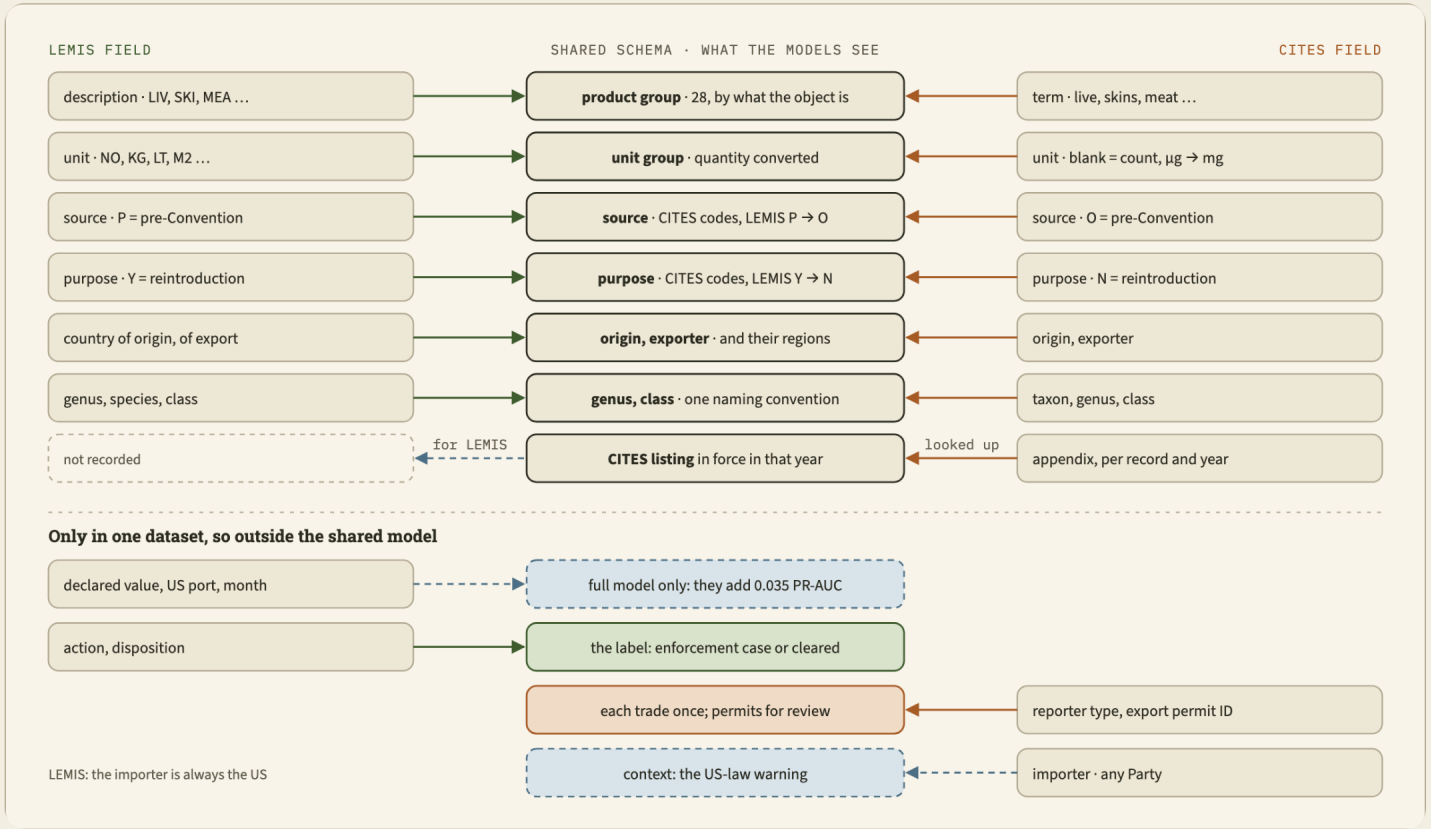
- Every US wildlife import declaration, 2000–2014
- 5.1 million line items of animals with a class
- Each has an outcome: cleared, or refused, seized, abandoned, re-exported
- One country only

CITES Trade Database

- Trade in CITES-listed species reported by every Party
- 8.2 million animal records, 2015–2024
- No enforcement outcomes
- Global

WildGuard learns from LEMIS outcomes and ranks CITES records: **cross-dataset label transfer**, judged by six research questions, from alignment (RQ1) to plausibility on CITES (RQ6).

Two datasets, one shared schema



Codes that look alike are not: pre-Convention is P in LEMIS but O in CITES; reintroduction is Y against N.

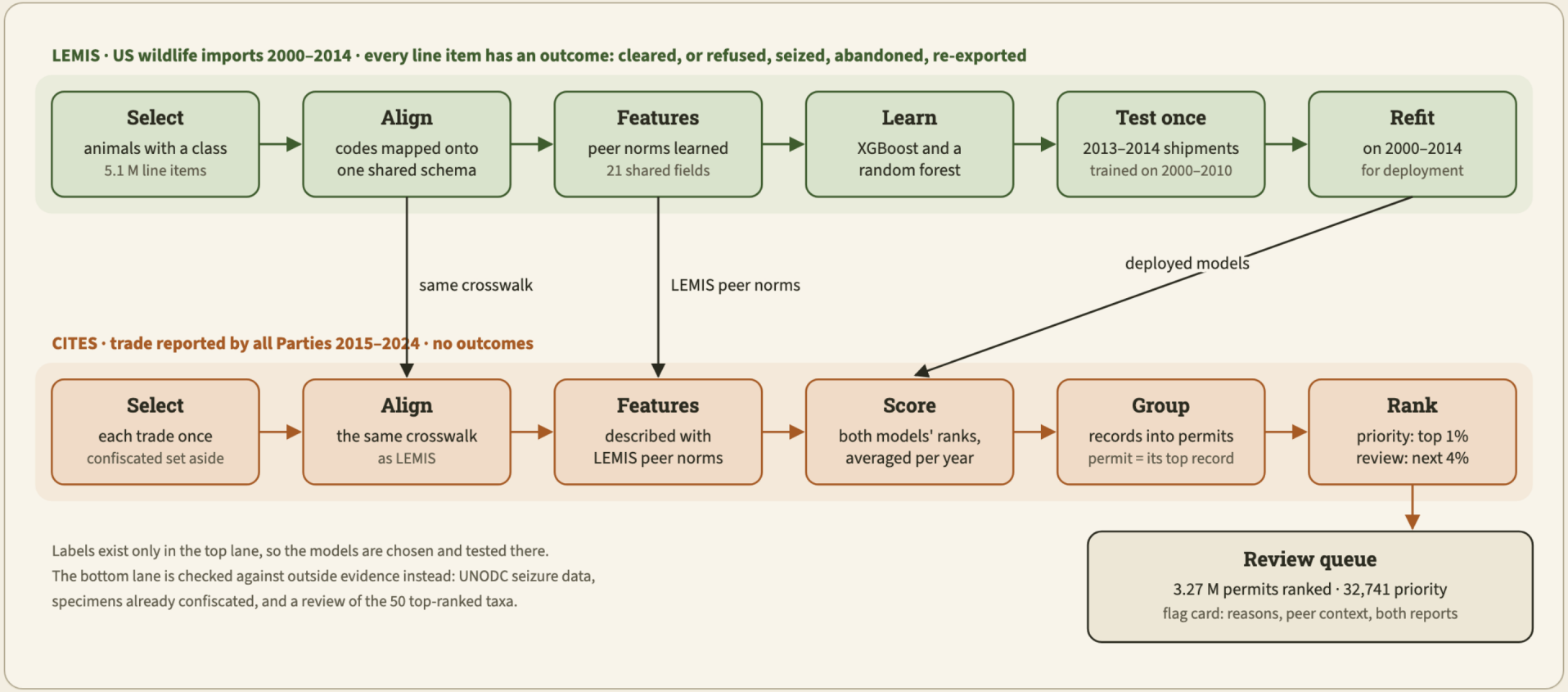
Products are grouped by what the object is, not by matching names.

The CITES listing in force in each shipment year is looked up for LEMIS.

What cannot cross (value, port, month) costs 0.035 PR-AUC.

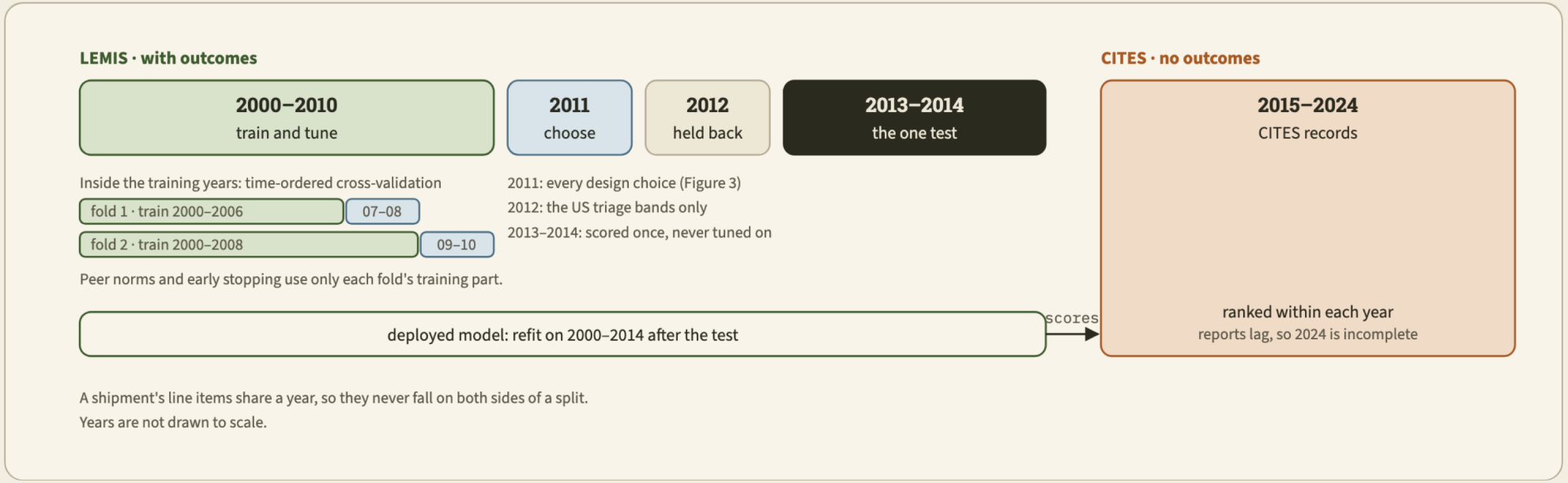
HOW IT WORKS · ARCHITECTURE

Learn where outcomes exist, rank where they do not



HOW IT IS EVALUATED

Nothing is tuned on the years it is judged on

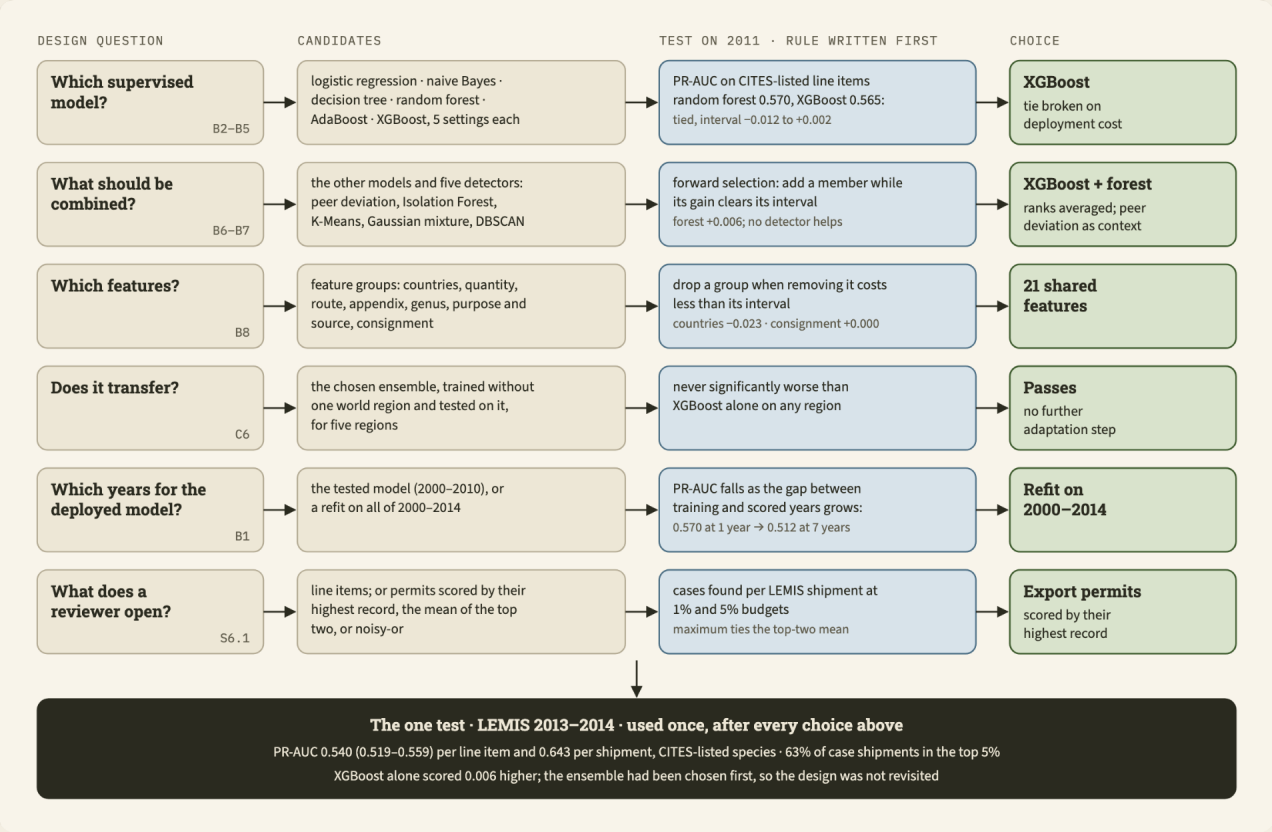


PR-AUC on CITES-listed line items decides: cases are under 2%, so accuracy says nothing

Effort per shipment, the unit an inspector opens: cases found at 1%, 5% and 10% budgets

95% intervals from a bootstrap of whole shipments; baselines: random and “most endangered first”

Every model choice made by comparison



Six supervised candidates, from logistic regression to XGBoost, and **five unsupervised detectors**, all under one protocol.

Each **rule was written before its result**; ties go to the simpler option.

One choice came after a result: the random forest–XGBoost tie was broken on **deployment cost**.

The deployed score: **XGBoost + random forest**, ranks averaged.

DOES IT WORK? (RQ2)

The one test: US imports 2013–2014, never seen

0.540

PR-AUC per line item on CITES-listed species, 27 times random (95% interval 0.519–0.559)

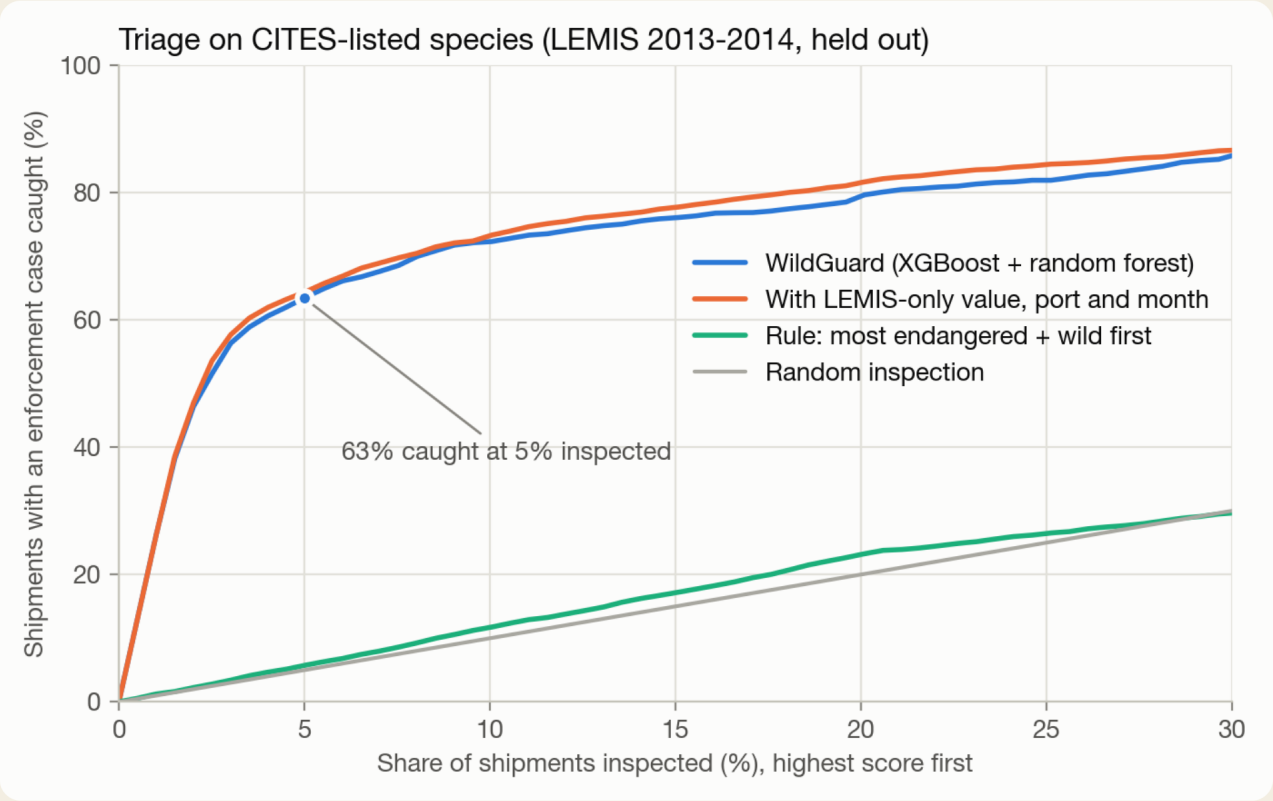
0.643 per shipment · listing rule 0.026 · random 0.020

Model (CITES-listed species)	Per line item	Per shipment
WildGuard: XGBoost + forest	0.540	0.643
XGBoost alone	0.546	0.649
Random forest alone	0.509	0.610
Logistic regression	0.470	0.567
Listing rule (most endangered first)	0.026	0.039
Random inspection	0.020	0.035

> XGBoost alone scored 0.006 higher (interval −0.009 to −0.003). The ensemble was chosen first, on 2011, and kept, so the test stays a test.

IS IT USEFUL? (RQ3)

What an inspection budget buys



63%

of shipments with a case found by inspecting 5% (random 5%, listing rule 6%)

21%

of shipments inspected to find 80% of the cases, instead of 80%

99%

of the top 1% of shipments had a case (base rate 3.8%)

COMBINING EVIDENCE AND TRANSFER (RQ4, RQ5)

What helped, and what did not

No detector helped

Five unsupervised detectors rank cases above random, but none improved the models. Unusual is not illegal.

Two models, small gain

Adding a random forest to XGBoost gained 0.006 on 2011, and lost 0.006 on the test.

Class weights hurt

Reweighting lowered PR-AUC for every strong candidate: it moves a threshold, not the ranking.

It transfers across regions

Trained without a world region and tested on it, the ensemble held up on all five.

But it ages

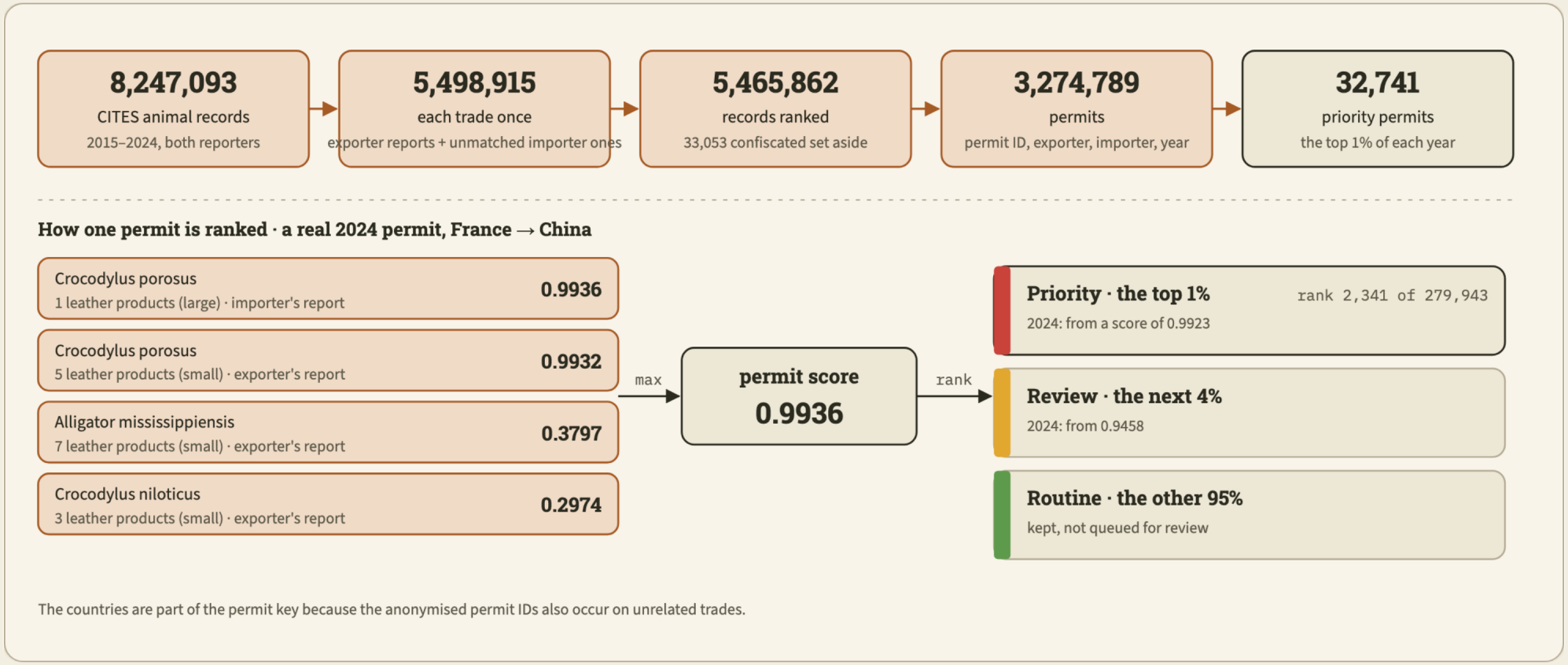
PR-AUC 0.570 one year after training, 0.512 seven years after, so it is refit on 2000–2014.

The shift is large

A classifier tells LEMIS from CITES with AUC 0.985; jurisdiction cannot be measured at all.

WHAT IT SURFACES IN CITES (RQ6)

From CITES records to a review queue



IS IT PLAUSIBLE? (RQ6) · PASS MARKS FIXED IN ADVANCE

Checked against outside evidence: one pass, two fails

FAIL

UNODC seizure data

Groups seized more than their legal trade suggests are not clearly flagged more: ρ = 0.13 (p = 0.32), below the listing rule's 0.43.

PASS

Confiscated specimens

Retrained without any source field, the model still ranks CITES's confiscated specimens far above the rest: ROC-AUC 0.888.

FAIL

The top 50 taxa

62% show a recognised risk pattern, but 30% none (pass: at most 20%), all personal imports of wild Appendix II specimens.

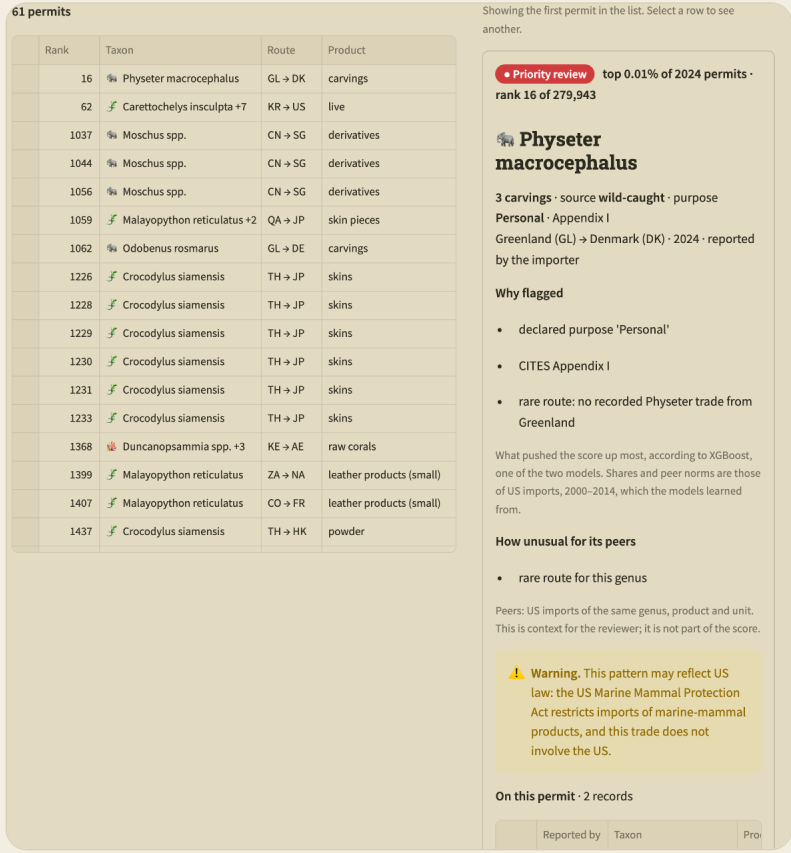
Half the priority tier is Siamese crocodile leather from Thailand: lawful CITES trade that US law prohibits (95% of LEMIS imports refused). WildGuard ranks trade by how US enforcement would treat it.

DEMO · FOR A CITES OR CUSTOMS REVIEWER

The review queue

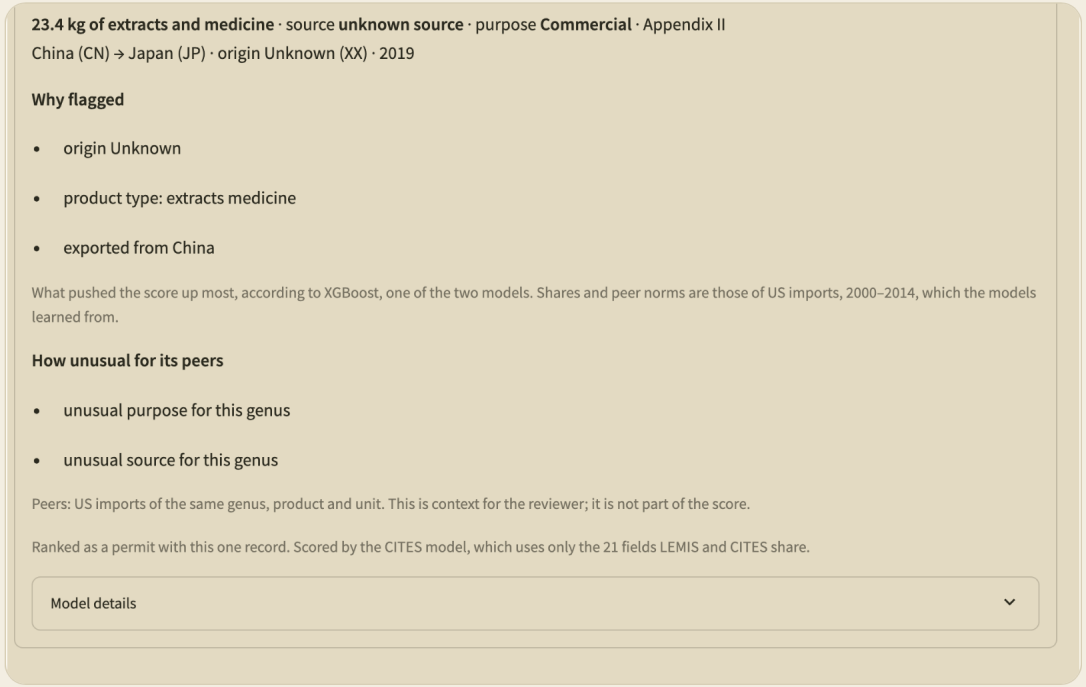
1. Open the queue: this year's priority permits, the top 1%, come first
2. Select a permit: its tier and its rank in the year
3. Read why it was flagged, and how unusual it is for its peers
4. Compare the permit's records, both countries' reports side by side
5. Heed the warning where US law, not risk, may explain the flag

Shown: sperm-whale carvings, Greenland → Denmark, 2024. The two reports disagree (carvings against teeth), and the US-law warning applies.

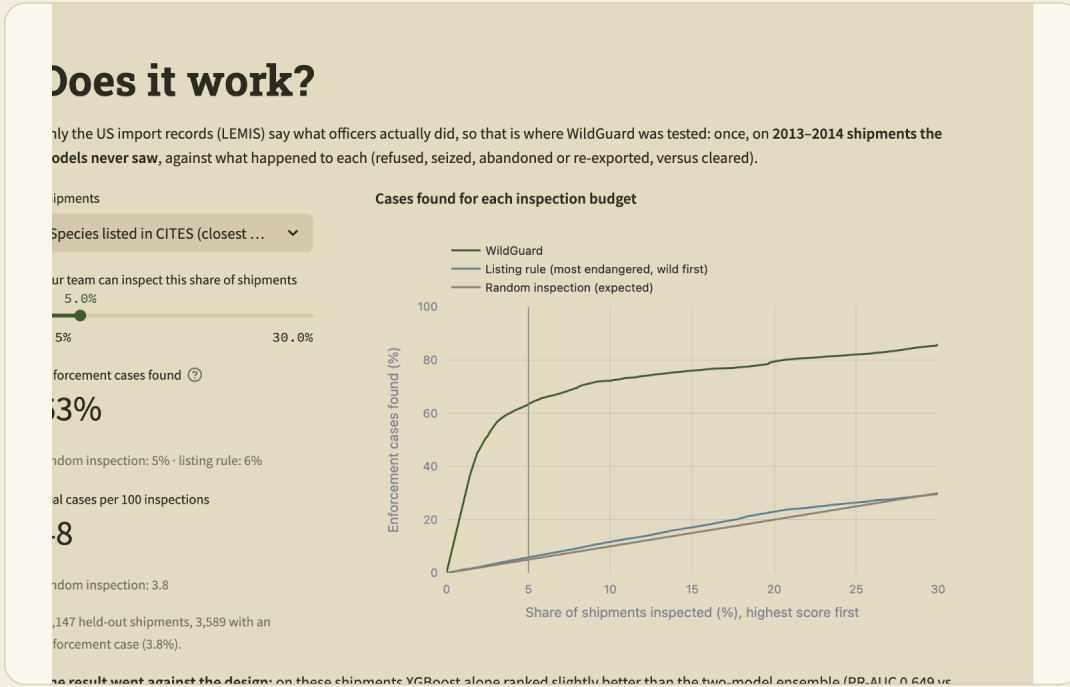


DEMO · CHECKING AND TRUSTING A FLAG

Check any record, see the evidence



Check a record: saiga-antelope medicine, China → Japan, ranks in the top 0.01% of 2019 permits. A US declaration can add value, port and month.



Does it work? Move the inspection budget and see the cases found on held-out US shipments, against random and the listing rule.

RESPONSIBLE USE

Limits, stated plainly

- **A flag is a prompt, not proof.** Labels are US enforcement outcomes, and most stopped shipments never reach CITES at all.
- **US law travels with the labels:** Siamese crocodile leather and marine mammals; only the latter carries a warning.
- **Commercial trade is harder:** PR-AUC 0.258 on commercial imports against 0.597 overall, and CITES is 90% commercial.
- **Old labels:** LEMIS ends in 2014, and the ranking ages with the gap to the scored year.
- **No accuracy claim on CITES:** two of the three checks fail; scores are rankings, never probabilities.
- **Permits exist only in CITES,** so signals such as the two reports disagreeing cannot be learned, only shown.

CONCLUSION

What WildGuard shows

Enforcement knowledge partly transfers. On US imports it never saw, inspecting 5% of shipments finds 63% of the cases, against 6% for the obvious manual rule.

On CITES it shows its origin. It finds confiscated specimens it was never told about, but not the UNODC seizure pattern, and much of what it flags is trade US law prohibits.

So it is a triage aid, not a verdict: every flag carries its reasons and its warnings, and a person decides.

Next: a labelled sample from a CITES authority · warnings for every US prohibition · a rebuild with each CITES release